

Mid-term examination on Prolog

Christian Rinderknecht

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1 Matching

Question. Show the results (Yes/No) and resulting variable bindings for the queries

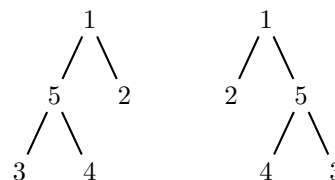
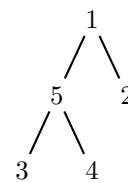
1. $?- f(g(X,X), h(Y,Y)) = f(g(Z), Z).$
2. $?- f(g(X,X), h(Y,Y)) = f(g(h(W,a),Z), Z).$
3. $?- f(g(X,X), h(_, _)) = f(g(h(W,a),Z), Z).$
4. $?- f(x(A,B), C) = f(C, x(B,A)).$

2 Mirroring binary trees

A *binary tree* is either a *leaf* or is made of a *root* and two subtrees (left and right). Here roots and leaves are integers. Assume the predicates `leaf(X)`, which denotes a tree made of a leaf `X`, and `tree(Root, Left, Right)`, which denotes a tree made of a root `Root`, a left subtree `Left` and a right subtree `Right`.

Question. Define `mirror/2` such that `mirror(T1,T2)` means “Binary tree `T1` is the mirror of binary tree `T2`.” As an example, the following trees are mirrors of each other:

Consider `tree(1,tree(5,leaf(3),leaf(4)),leaf(2))` denoting



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?- mirror(tree(1,tree(5,leaf(3),leaf(4)),leaf(2)), Mirror).
Mirror = tree(1, leaf(2), tree(5, leaf(4), leaf(3))) ;
No
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